

Title

Mechanisms and moderators of smartphone-based cessation with varying therapeutic intensity: subgroup and longitudinal analyses from a randomized smoking intervention trial

Abstract

Understanding which smokers benefit most from intensive therapeutic contact versus self-guided app-based approaches could enable more personalized cessation interventions. Secondary analyses examined 420 trial participants randomized to app-only, app-plus-high-intensity-support (eight weekly sessions), or app-plus-low-intensity-support (four biweekly sessions). We conducted pre-specified subgroup analyses stratified by baseline motivation, cigarettes-per-day, number of prior-quit attempts, age, and baseline mental-health status. High-intensity therapeutic contact demonstrated consistent benefit across most demographic and clinical subgroups, with notably larger absolute effect sizes in heavy smokers and those with multiple prior-quit attempts. App-only participants with high baseline motivation achieved cessation rates approximating those of low-intensity-supported participants. Engagement metrics revealed that app-usage frequency correlated with abstinence success in all arms but was highest among participants in therapeutic-contact conditions. Qualitative exit interviews identified social accountability and structured therapeutic check-ins as key perceived mechanisms of benefit. These secondary findings suggest selective intensification of therapeutic contact for specific populations could optimize resource allocation in scaled smoking-cessation programs.